2. **Microcalorimeter** (9 points)

1) Every bridge has thermal resistance $L/\kappa S$; so, the overall resistance is $R = L/4\kappa S$.

2) The power dissipation P results in an heat flux through the bridges, $\Phi = \Delta T/R$, and in the change of the heat contained in the microcalorimeter, $\dot{Q} = C\dot{T} = C\Delta\dot{T}$ (here, dot denotes the time derivative). So,

$$P_0 \cos(\omega t) = C\Delta\dot{T} + \Delta T/R.$$

Now, we can search the solution as $\Delta T = A\cos(\omega t + \phi)$, and denote $\psi = \arcsin(C\omega/\sqrt{C^2\omega^2 + R^{-2}})$. Then,

$$P_0 \cos(\omega t) = A\sqrt{C^2\omega^2 + R^{-2}}\cos(\omega t + \phi - \psi).$$

So, we must have $\phi = \psi$ and $A = P_0/\sqrt{C^2\omega^2 + R^{-2}}$, i.e.

$$T = T_0 + \frac{P_0 \cos\left(\omega t + \arcsin(C\omega/\sqrt{C^2\omega^2 + R^{-2}})\right)}{\sqrt{C^2\omega^2 + R^{-2}}}.$$

3) The amplitude of the oscillations is $A = \frac{P_0}{\sqrt{C^2\omega^2 + R^{-2}}}$; it must be as sensitive with respect to the small changes of C, i.e. dA/dC must be maximal by modulus. $dA/dC = P_0(C^2\omega^2 + R^{-2})^{-3/2}C\omega^2$; if we denote $x = (C\omega)^2$, we need to minimize the following function of x:

$$\ln\left[(dC/dA)^2\right] = 3\ln(x + R^{-2}) - \ln x + \ln C.$$

Upon taking derivative and putting it equal to 0, we obtain $3x = x + R^{-2}$, from where $x = R^{-2}/2$, i.e.

$$\omega = 1/\sqrt{2}CR.$$

4) The heat contained in the bridge must be comparable with the heat, which flows through it during one half-period (if it is much smaller, the stationary linear profile will develop very soon). So, $Ac\rho SL \approx A\kappa S/(L\omega_c)$; hence,

$$\omega_c \approx \kappa/c\rho L^2.$$